MTH 201 – Business Calculus: Midterm (21/4/2026)

Full name: ..... **TROY ID No.:** .....

Q1. Find any relative extrema of $f(x) = e^x(x-3)$. State where the function is increasing or decreasing and determine points of inflection.

Q2. Find the absolute maximum and minimum values of the function $f(x) = \frac{x}{9x^2+1}$ over $[0;2]$.

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Q3. The total cost and total revenue functions for a company are

$$C(x) = x^3 - 22x^2 + 96x + 310,\ R(x) = x(171 - 2x).$$

a) Write down the equation for the profit function $P(x)$. Hence determine the level of output, x, for which profit is maximized.

b) Find the marginal revenue $MR(x)$ and the marginal cost $MC(x)$. Verify that $MR = MC$ when the profit function has a maximum value.

Q4. A rectangular box with a square base and a cover is to have a volume of 875 cm^3. If the cost per square cm for the bottom is \$4, for the top is \$3, and for the sides is \$4, what should the dimensions be in order to minimize the cost?